

# PROJECT RISK MANAGEMENT

# Overview

- Risk
  - An **uncertain** event or condition that, if it occurs, has a positive or negative effect on the project objectives.
- Project Risk Management includes the processes of conducting risk management planning, identification, analysis, response planning, response implementation, and monitoring risk on a project.
- The objectives of project risk management are **to increase the probability and/or impact of positive risks** and to **decrease the probability and/or impact of negative risks**, in order to optimize the chances of project success.

# Risk Management Process

- **11.1 Plan Risk Management**—The process of defining **how to conduct** risk management activities for a project. **11.2 Identify Risks**—The process of identifying individual project risks as well as sources of overall project risk,
  - and documenting their characteristics.
- **11.3 Perform Qualitative Risk Analysis**—The process of **prioritizing** individual project risks for further analysis or action by assessing their **probability of occurrence and impact** as well as other characteristics.
- **11.4 Perform Quantitative Risk Analysis**—The process of **numerically** analyzing the combined effect of identified individual project risks and other sources of uncertainty on overall project objectives.
- **11.5 Plan Risk Responses**—The process of **developing options**, selecting strategies, and agreeing on actions to address overall project risk exposure, as well as to treat individual project risks.
- **11.6 Implement Risk Responses**—The process of **implementing** agreed-upon risk response plans.
- **11.7 Monitor Risks**—The process of **monitoring** the implementation of agreed-upon risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating risk process effectiveness throughout the project.

## Project Risk Management Overview

### 11.1 Plan Risk Management

- .1 Inputs
  - .1 Project charter
  - .2 Project management plan
  - .3 Project documents
  - .4 Enterprise environmental factors
  - .5 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Data analysis
  - .3 Meetings
- .3 Outputs
  - .1 Risk management plan

### 11.5 Plan Risk Responses

- 1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Enterprise environmental factors
  - .4 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Data gathering
  - .3 Interpersonal and team skills
  - .4 Strategies for threats
  - .5 Strategies for opportunities
  - .6 Contingent response strategies
  - .7 Strategies for overall project risk
  - .8 Data analysis
  - .9 Decision making
- .3 Outputs
  - .1 Change requests
  - .2 Project management plan updates
  - .3 Project documents updates

### 11.2 Identify Risks

- .1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Agreements
  - .4 Procurement documentation
  - .5 Enterprise environmental factors
  - .6 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Data gathering
  - .3 Data analysis
  - .4 Interpersonal and team skills
  - .5 Prompt lists
  - .6 Meetings
- .3 Outputs
  - .1 Risk register
  - .2 Risk report
  - .3 Project documents updates

### 11.6 Implement Risk Responses

- .1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Interpersonal and team skills
  - .3 Project management information system
- .3 Outputs
  - .1 Change requests
  - .2 Project documents updates

### 11.3 Perform Qualitative Risk Analysis

- .1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Enterprise environmental factors
  - .4 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Data gathering
  - .3 Data analysis
  - .4 Interpersonal and team skills
  - .5 Risk categorization
  - .6 Data representation
  - .7 Meetings
- .3 Outputs
  - .1 Project documents updates

### 11.7 Monitor Risks

- .1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Work performance data
  - .4 Work performance reports
- .2 Tools & Techniques
  - .1 Data analysis
  - .2 Audits
  - .3 Meetings
- .3 Outputs
  - .1 Work performance information
  - .2 Change requests
  - .3 Project management plan updates
  - .4 Project documents updates
  - .5 Organizational process assets updates

### 11.4 Perform Quantitative Risk Analysis

- .1 Inputs
  - .1 Project management plan
  - .2 Project documents
  - .3 Enterprise environmental factors
  - .4 Organizational process assets
- .2 Tools & Techniques
  - .1 Expert judgment
  - .2 Data gathering
  - .3 Interpersonal and team skills
  - .4 Representations of uncertainty
  - .5 Data analysis
- .3 Outputs
  - .1 Project documents updates

# Managing Risks

- Risks will continue to emerge during the lifetime of the project, so Project Risk Management processes should be conducted **iteratively**. Risk is initially addressed during project planning by shaping the project strategy. Risk should also be monitored and managed as the project progresses to ensure that the project stays on track and emergent risks are addressed

# Risk factors

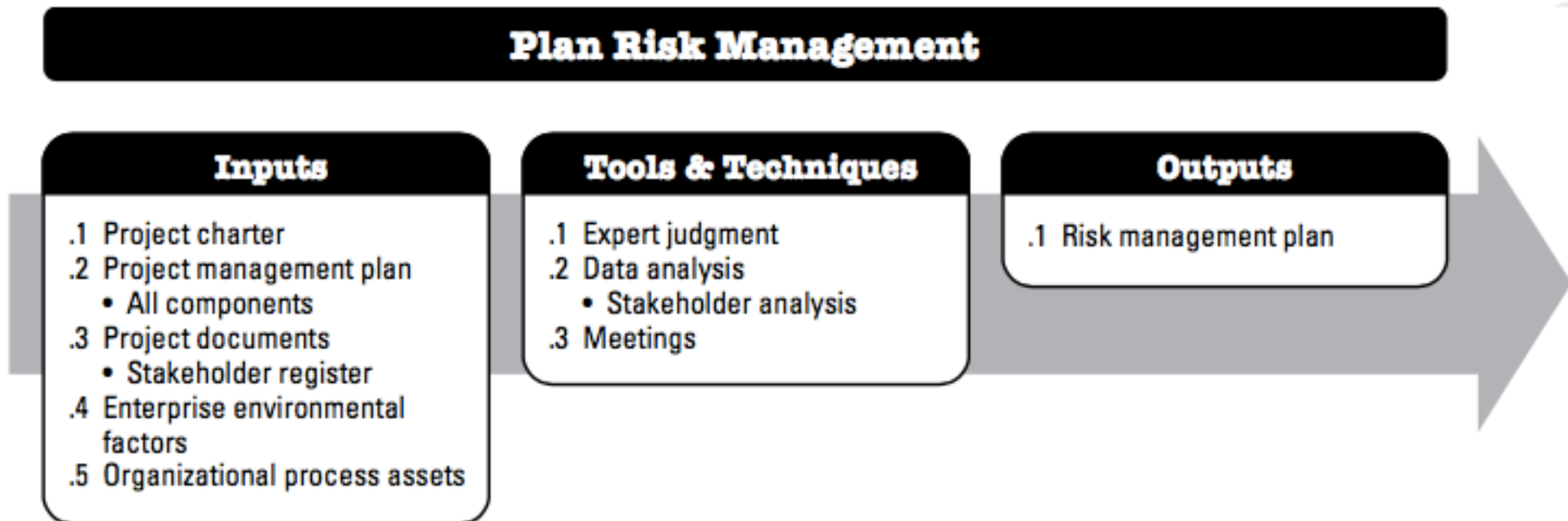
- Probability
  - Impact
  - Expected timing
  - Frequency of the event
- 
- Risk management process helps prevent future problems from occurring

# Tailoring Consideration

- Because each project is unique, it is necessary to tailor the way Project Risk Management processes are applied. Considerations for tailoring include but are not limited to:
- **◆ Project size.** Does the project's size in terms of budget, duration, scope, or team size require a more detailed approach to risk management? Or is it small enough to justify a simplified risk process?
- **◆ Project complexity.** Is a robust risk approach demanded by high levels of innovation, new technology, commercial arrangements, interfaces, or external dependencies that increase project complexity? Or is the project simple enough that a reduced risk process will suffice?
- **◆ Project importance.** How strategically important is the project? Is the level of risk increased for this project because it aims to produce breakthrough opportunities, addresses significant blocks to organizational performance, or involves major product innovation?
- **◆ Development approach.** Is this a waterfall project, where risk processes can be followed sequentially and iteratively, or does the project follow an agile approach where risk is addressed at the start of each iteration as well as during its execution?
- Tailoring of the Project Risk Management processes to meet these considerations is part of the Plan Risk Management process, and the outcomes of tailoring decisions are recorded in the risk management plan.

# Plan Risk Management

- Plan Risk Management is the process of defining how to conduct risk management activities for a project. The key benefit of this process is that it ensures that the degree, type, and visibility of risk management are proportionate to both risks and the importance of the project to the organization and other stakeholders.





# Risk Management Plan

- The risk management plan is a component of the project management plan that describes how risk management activities will be structured and performed. The risk management plan may include some or all of the following elements:
- **Risk strategy.** Describes the general approach to managing risk on this project.
- **Methodology.** Defines the specific approaches, tools, and data sources that will be used to perform risk management on the project.
- **Roles and responsibilities.** Defines the lead, support, and risk management team members for each type of activity described in the risk management plan, and clarifies their responsibilities.
- **Funding.** Identifies the funds needed to perform activities related to Project Risk Management. Establishes protocols for the application of contingency and management reserves.
- **Timing.** Defines when and how often the Project Risk Management processes will be performed throughout the project life cycle, and establishes risk management activities for inclusion into the project schedule.
- **Risk categories.** Provide a means for grouping individual project risks. A common way to structure risk categories is with a risk breakdown structure (RBS), which is a hierarchical representation of potential sources of risk (see example in Figure 11-4). An RBS helps the project team consider the full range of sources from which individual project risks may arise. This can be useful when identifying risks or when categorizing identified risks. The organization may have a generic RBS to be used for all projects, or there may be several RBS frameworks for different types of projects, or the project may develop a tailored RBS. Where an RBS is not used, an organization may use a custom risk categorization framework, which may take the form of a simple list of categories or a structure based on project objectives.

# Risk Breakdown Structure

| RBS LEVEL 0                    | RBS LEVEL 1        | RBS LEVEL 2                                 |
|--------------------------------|--------------------|---------------------------------------------|
| 0. ALL SOURCES OF PROJECT RISK | 1. TECHNICAL RISK  | 1.1 Scope definition                        |
|                                |                    | 1.2 Requirements definition                 |
|                                |                    | 1.3 Estimates, assumptions, and constraints |
|                                |                    | 1.4 Technical processes                     |
|                                |                    | 1.5 Technology                              |
|                                |                    | 1.6 Technical interfaces                    |
|                                |                    | Etc.                                        |
|                                | 2. MANAGEMENT RISK | 2.1 Project management                      |
|                                |                    | 2.2 Program/portfolio management            |
|                                |                    | 2.3 Operations management                   |
|                                |                    | 2.4 Organization                            |
|                                |                    | 2.5 Resourcing                              |
|                                |                    | 2.6 Communication                           |
|                                |                    | Etc.                                        |
|                                | 3. COMMERCIAL RISK | 3.1 Contractual terms and conditions        |
|                                |                    | 3.2 Internal procurement                    |
|                                |                    | 3.3 Suppliers and vendors                   |
|                                |                    | 3.4 Subcontracts                            |
|                                |                    | 3.5 Client/customer stability               |
|                                |                    | 3.6 Partnerships and joint ventures         |
|                                |                    | Etc.                                        |
|                                | 4. EXTERNAL RISK   | 4.1 Legislation                             |
|                                |                    | 4.2 Exchange rates                          |
|                                |                    | 4.3 Site/facilities                         |
|                                |                    | 4.4 Environmental/weather                   |
|                                |                    | 4.5 Competition                             |
|                                |                    | 4.6 Regulatory                              |
|                                |                    | Etc.                                        |

# Identify Risks

- Identify potential risks that can impact the project
  - Includes both threats and opportunities
- Should include many of the project stakeholders
- The IT Project Risk Framework provides a tool for understanding the timing and interrelatedness of IT project risks

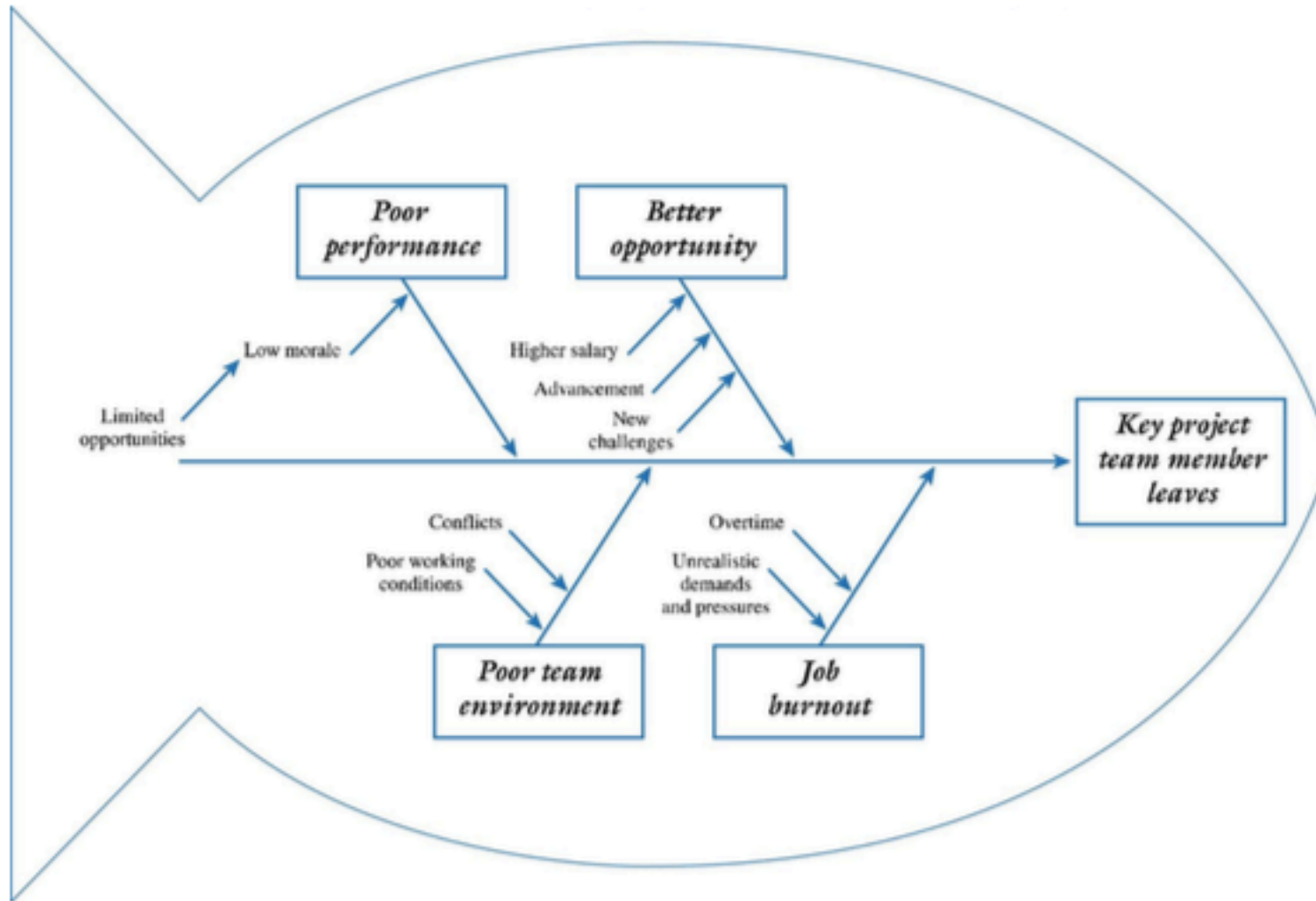
# Tools for Identifying Risks

- Learning Cycles
- Brainstorming
- Nominal Group Technique
- Delphi Technique
- Checklists
- SWOT Analysis
- Cause & Effect (a.k.a. Fishbone/Ishikawa)
- Past Projects

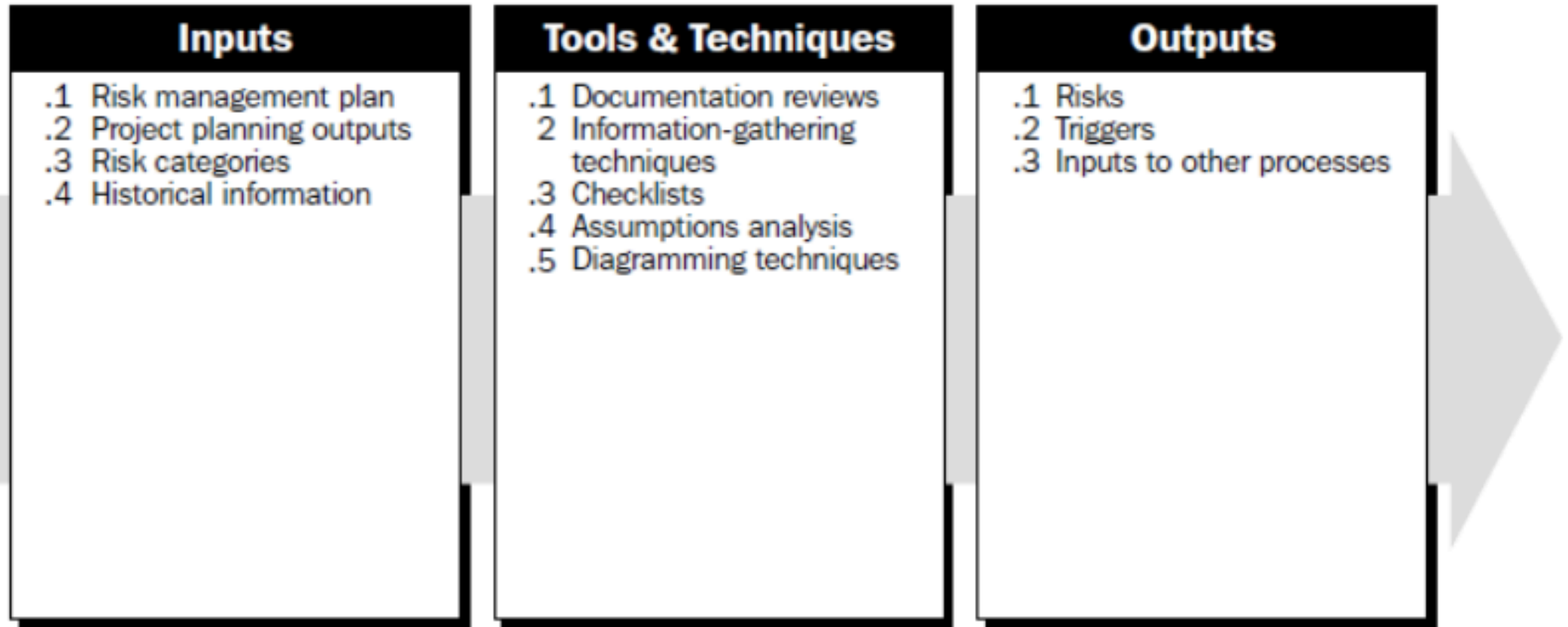
# Example of a Risk Checklist

- ☐ Funding for the project has been secured
  - ☐ Funding for the project is sufficient
  - ☐ Funding for the project has been approved by senior management
  - ☐ The project team has the requisite skills to complete the project
  - ☐ The project has adequate manpower to complete the project
- ☐ The project charter and project plan have been approved by senior management or the project sponsor
  - ☐ The project's goal is realistic and achievable
  - ☐ The project's schedule is realistic and achievable
  - ☐ The project's scope has been clearly defined ☐ Processes for scope changes have been clearly defined

# Cause and effect diagram



# Risk Identification



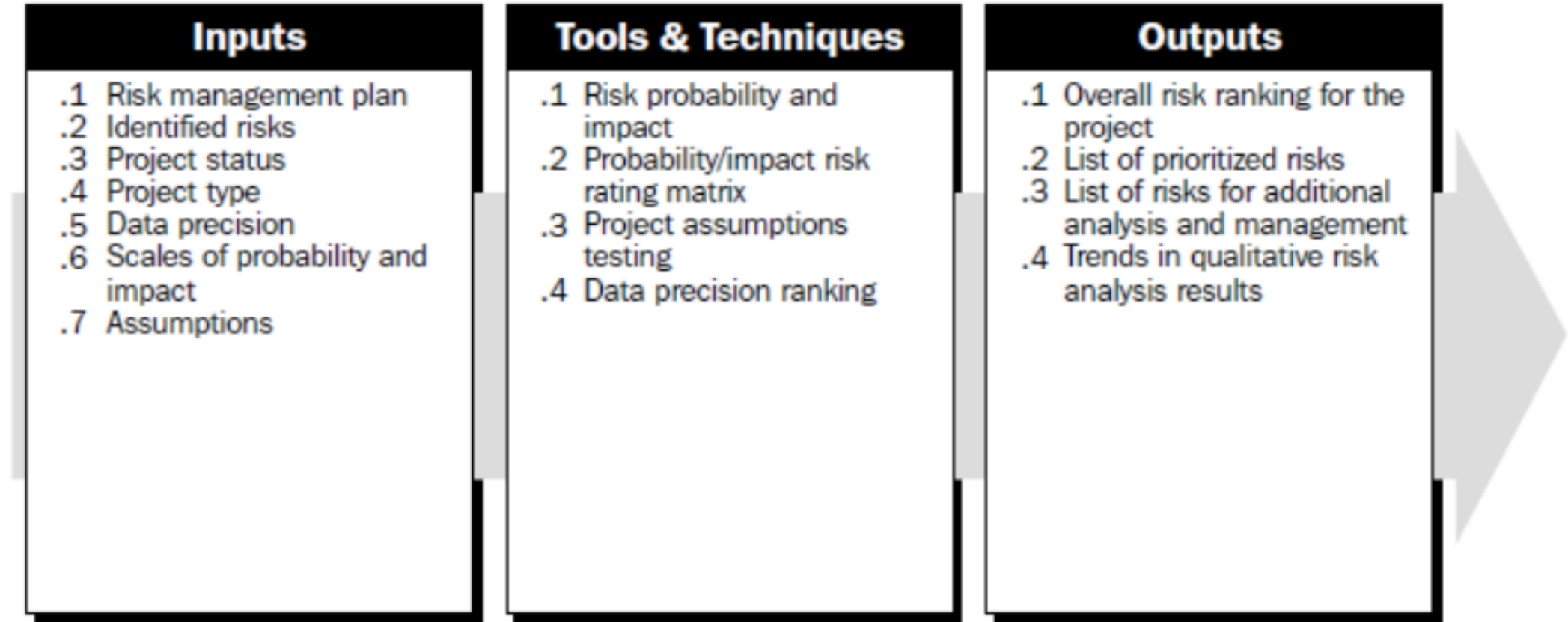
# How to Identify Risk

Business Risk:

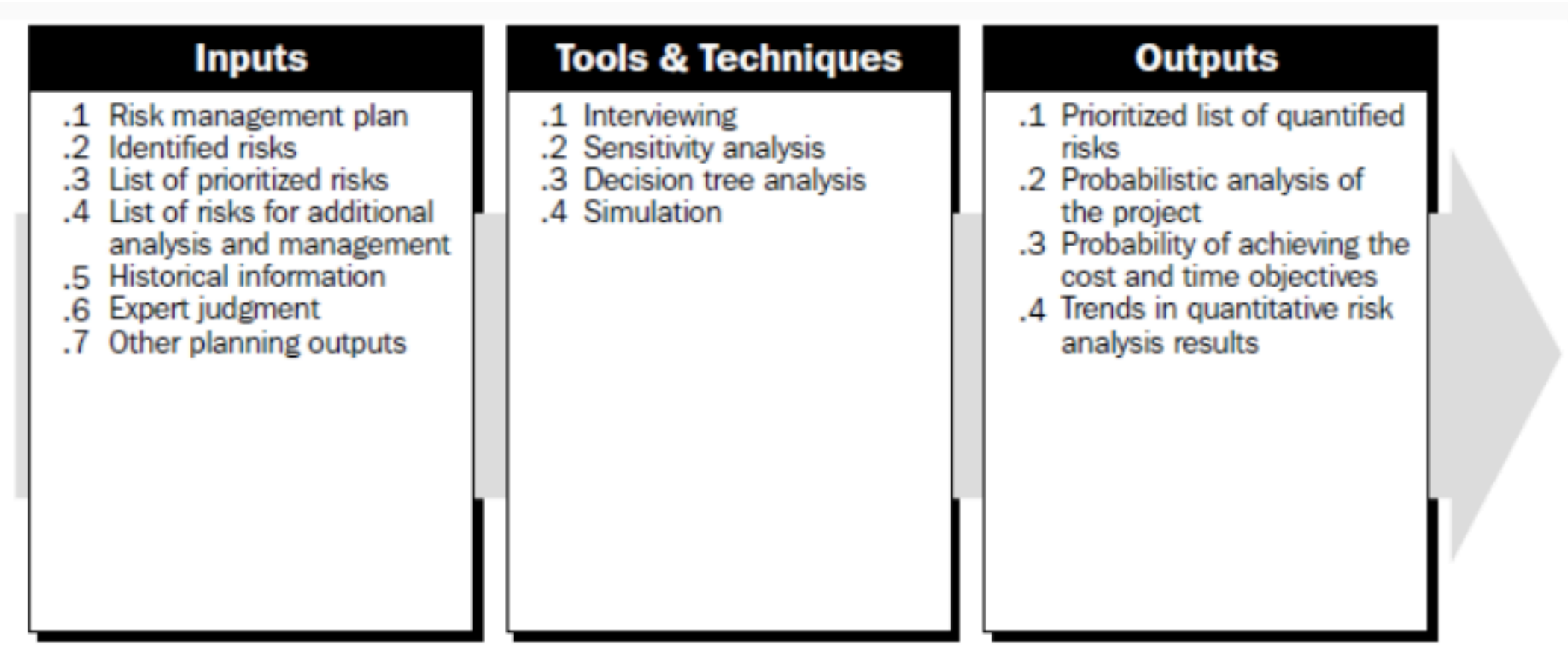
| Cause                                                 | Risk                                        | Impact                                                 |
|-------------------------------------------------------|---------------------------------------------|--------------------------------------------------------|
| Due to a change in governmental policies (regulation) | The project funding may be stopped          | Which may result in premature closeout of the contract |
| As a result of technological advancement              | The software currently used may be outdated | Resulting in new training requirements                 |



# Qualitative Risk Analysis



# Quantitative Risk Analysis

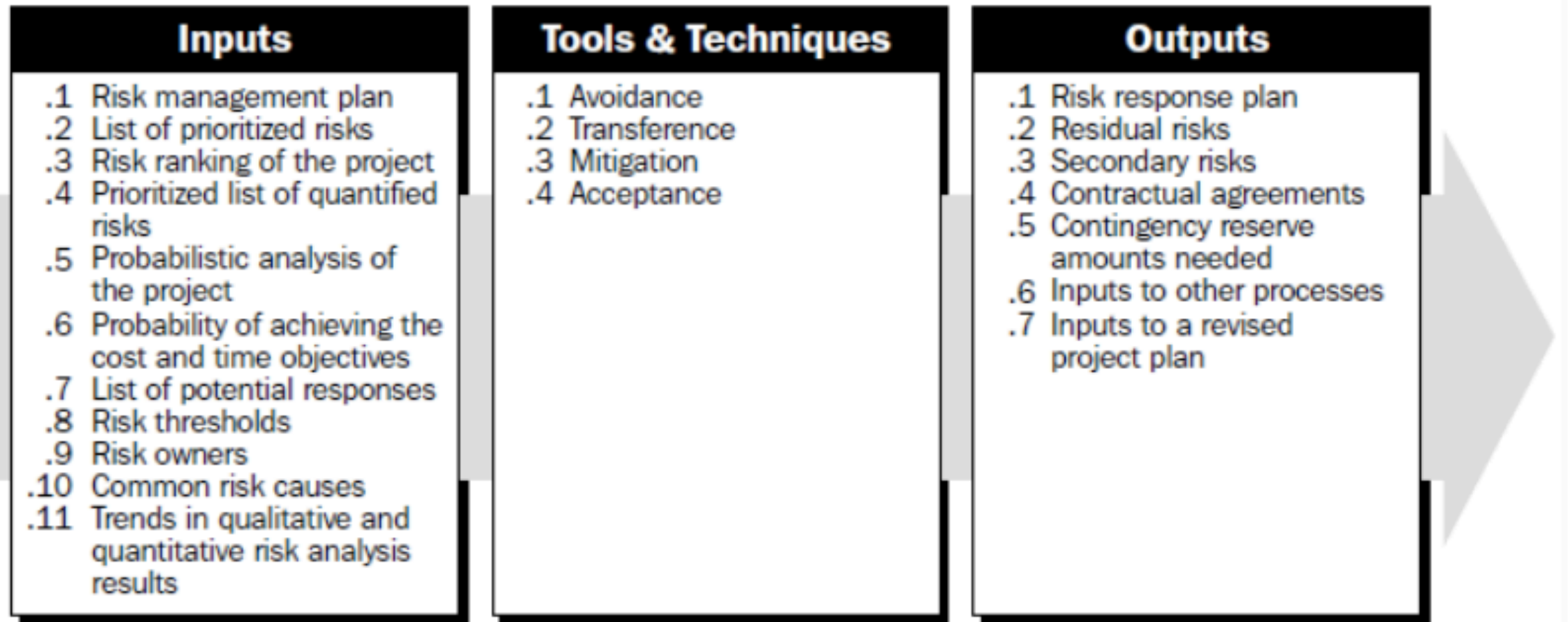


# Risk Strategy

Depends On:

- The nature of the risk itself
  - Really a threat or an opportunity?
- The impact of the risk on the project's MOV and objectives
  - What is the probability and impact of a risk
- The project's constraints in terms of scope, schedule, budget, and quality
  - Can a response be made with existing resources and/or constraints?
- Risk Tolerances or preferences of the project stakeholders
  - How much risk is tolerable?

# Risk Response Planning



# Risk Monitoring and Control

